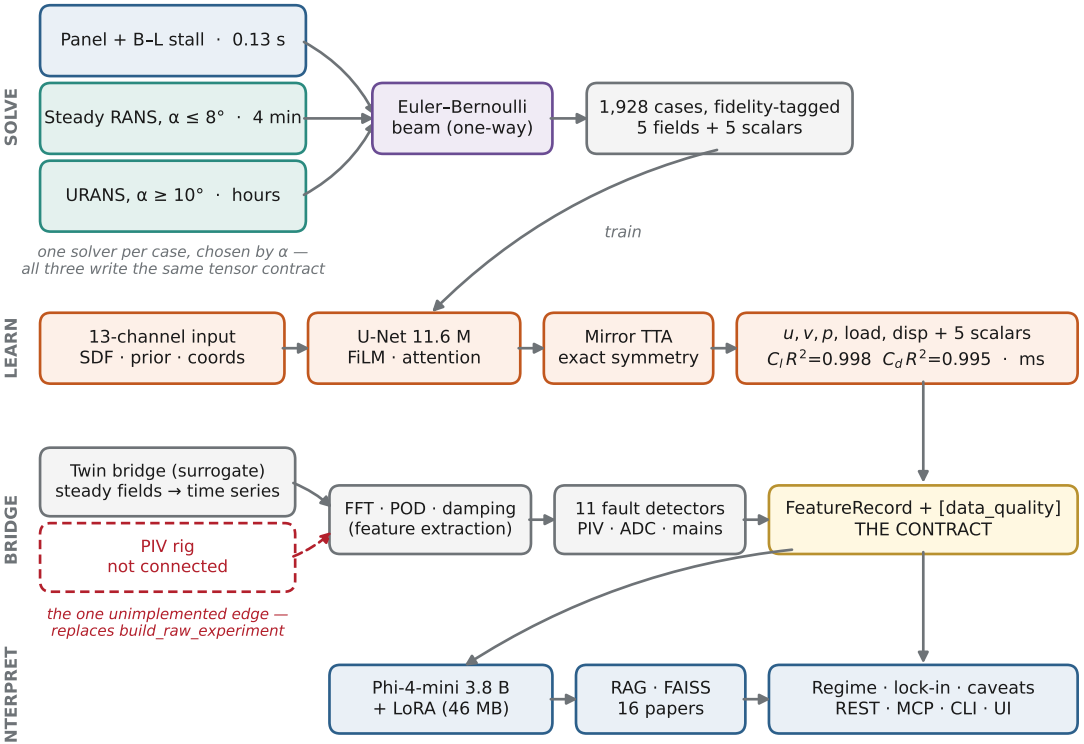
SOLVE
Panel + B–L stall · 0.13 s
Steady RANS, α ≤ 8° · 4 min
URANS, α ≥ 10° · hours
Euler–Bernoulli beam (one-way)
1,928 cases, fidelity-tagged
5 fields + 5 scalars
one solver per case, chosen by α —
all three write the same tensor contract
train
LEARN
13-channel input
SDF · prior · coords
U-Net 11.6 M
FiLM · attention
Mirror TTA
exact symmetry
u, v, p, load, disp + 5 scalars
$C_l R^2$=0.998 $C_d R^2$=0.995 · ms
BRIDGE
Twin bridge (surrogate)
steady fields → time series
PIV rig
not connected
FFT · POD · damping
(feature extraction)
11 fault detectors
PIV · ADC · mains
FeatureRecord + [data_quality]
THE CONTRACT
the one unimplemented edge —
replaces build_raw_experiment
INTERPRET
Phi-4-mini 3.8 B
+ LoRA (46 MB)
RAG · FAISS
16 papers
Regime · lock-in · caveats
REST · MCP · CLI · UI